

Cyber and Information Security

Lecture 6: Firewall



Learning objectives

01

Explain firewall concepts,

02

Explain firewall actions and types,

03

Explain the attacks on packet filtering firewalls and the appropriate countermeasures,

04

Explain firewall deployment architecture,

05

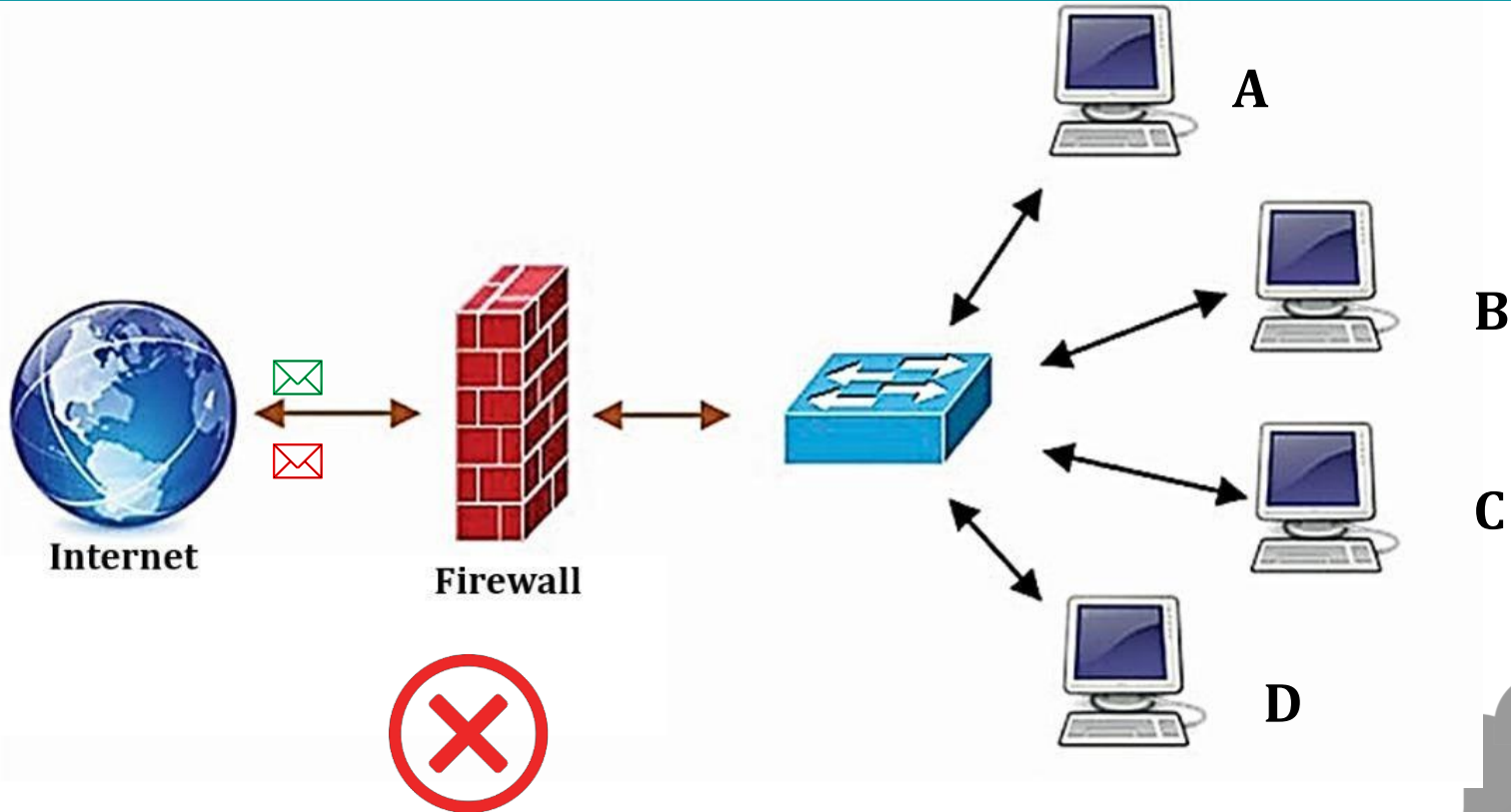
Understand firewall policies.

Network Firewall

- A firewall is a part of a computer system or network designed to stop unauthorized traffic from flowing from one network to another.
- In firewall, all inbound and outbound network traffic will be checked by the firewall.
- If the firewall identifies packets as legitimate packets, then it allows them to go to the other side of the firewall otherwise the firewall will not allow the packet to the intended destination.



Network Firewall (cont.)





Firewall Actions

IPv4 Firewall IPv6 Firewall

Hide empty chains Reset Counters Restart Firewall

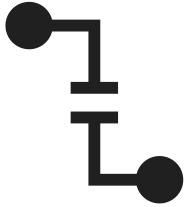
Table: Filter

Chain INPUT (Policy: ACCEPT)

Pkts.	Traffic	Target	Out	Source	Destination	State	Log	Chain
36.84 K	3.26 MB	ACCEPT	lo	*	0.0.0.0/0	0.0.0.0/0	-	-
4.75 M	1.56 GB	input_rule	all	*	0.0.0.0/0	0.0.0.0/0	-	Custom input rule chain
3.80 M	892.10 MB	ACCEPT	all	*	0.0.0.0/0	0.0.0.0/0	ctstate RELATED,ESTABLISHED	-
145.63 K	8.72 MB	syn_flood	tcp	*	0.0.0.0/0	0.0.0.0/0	tcp flags:0x17/0x02	-
372.08 K	332.92 MB	zone_lan_input	all	br-lan	0.0.0.0/0	0.0.0.0/0	-	-
14.84 K	874.72 KB	zone_wan_input	all	eth0.2	0.0.0.0/0	0.0.0.0/0	-	-
506.23 K	329.45 MB	zone_DMZ_input	all	br-DMZ	0.0.0.0/0	0.0.0.0/0	-	-
58.72 K	3.60 MB	zone_Kids_input	all	br-Kids	0.0.0.0/0	0.0.0.0/0	-	-

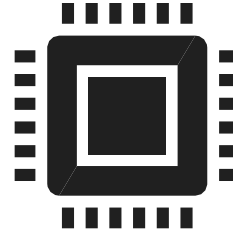


Firewall Actions (cont.)



Packets flowing through a firewall can have one of three outcomes:

- Accepted:** permitted through the firewall
- Denied:** not allowed through with no indication of failure
- Rejected:** discards but sends an error message back to the sender



Policies used by the firewall to handle packets abased on several properties of the packets being inspected, including the

Protocol used, such as TCP or UDP

The source and destination r@ IP addresses

The source and destination ports

The application-level payload of the packet.



Type of Filters in Firewall

Firewall actions rely on two types of filtering which are

Ingress filtering

Egress filtering



Type of Filters in Firewall (cont.)

Firewall actions rely on two types of filtering.

- ***Ingress filtering:*** *Inspects the incoming traffic to safeguard an internal network and prevent attacks from outside.*
- ***Egress filtering:*** *Inspects the outgoing network traffic and prevents the users in the internal network to reach out to the outside network.*
 - *For example, blocking social networking sites in school*

Blacklists and Whitelists

Two fundamental approaches to creating firewall policies (or

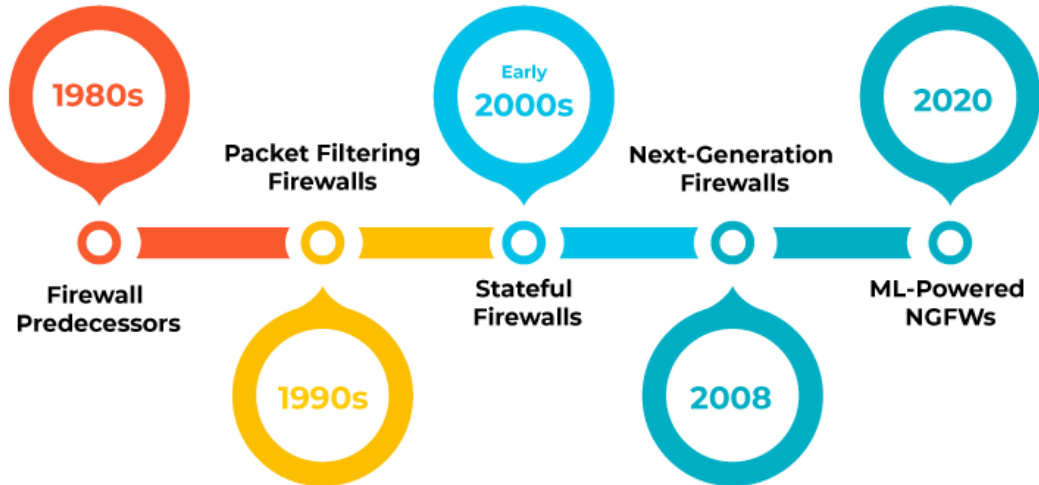
- **Blacklist** approach (default-allow)
 - All packets are allowed through except those that fit the rules defined specifically in a blacklist.
 - Pros: flexible in ensuring that service to the internal network is not disrupted by the firewall.
 - Cons: unexpected forms of malicious traffic could go through.
- **Whitelist** approach (default-deny)
 - Packets are dropped or rejected unless they are specifically allowed by the firewall.
 - Pros: A safer approach to defining a firewall ruleset.
 - Cons: must consider all possible legitimate traffic in rulesets.

Firewall Evolution

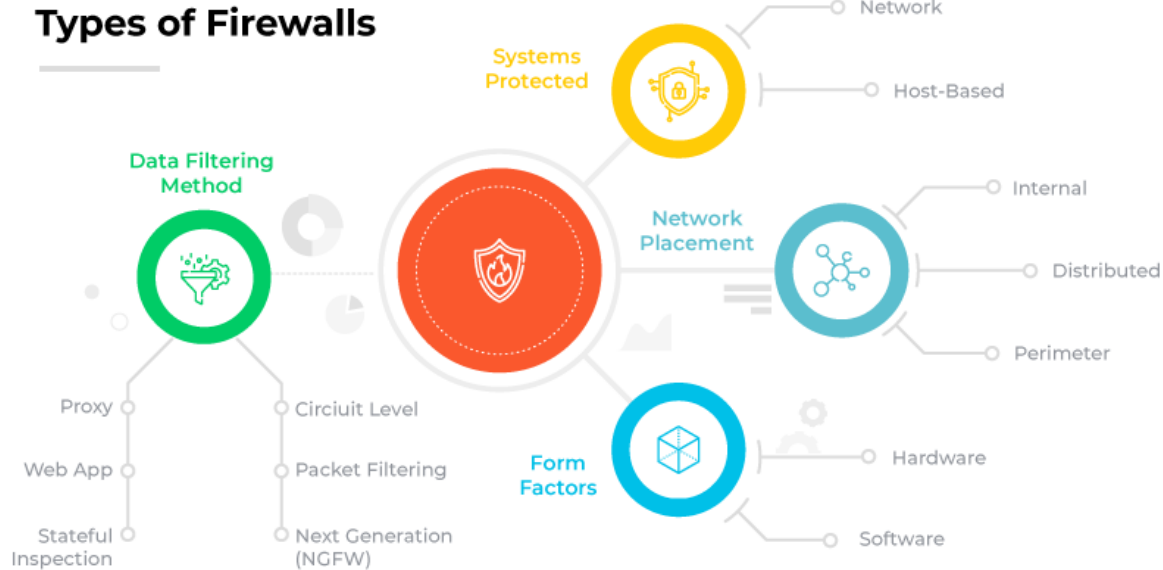
Generation	Era	Core Principle	Example
1st Gen – Packet Filtering	1980s	Filters packets by IP, port, protocol	Cisco IOS ACLs
2nd Gen – Stateful Inspection	1990s	Tracks connection states (session-aware)	Check Point Firewall-1
3rd Gen – Application/Proxy Firewall	2000s	Filters traffic by application or protocol	Squid Proxy
Next-Gen (NGFW)	2008s+	Deep packet inspection, IDS/IPS, app & user awareness	Palo Alto, Fortinet

Firewall Evolution

The History of Firewalls



What Are the Different Types of Firewalls?



Types of Firewalls According to Function

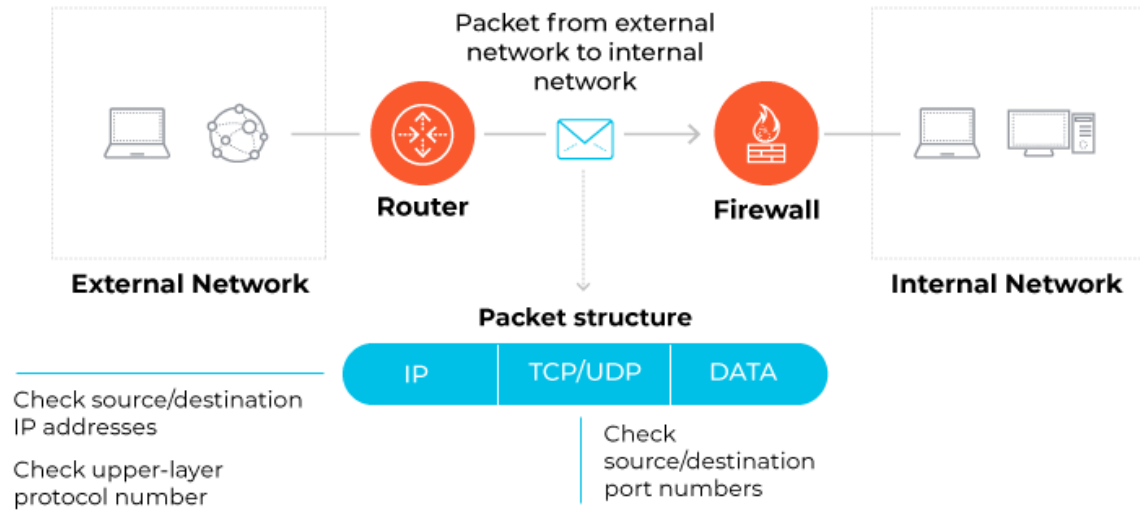
- Packet-filtering firewalls.
- Stateful inspection firewalls.
- Proxy firewalls.
- Web application firewalls (WAFs).
- Next-generation firewalls (NGFWs).
- Unified Threat Management (UTM) firewalls

Packet-filtering Firewalls

- A firewall that filters network packets based on a predefined set of rules.
- Rules are typically based on IP addresses, port numbers, and protocols.
- Firewall inspect packet headers, it decides
 - if it matches an allowed rule;
 - if not, it blocks the packet.
- It does not inspect packet contents for potential threats.

Packet-filtering Firewalls

How a Packet Filtering Firewall Works



What does data packet mean?



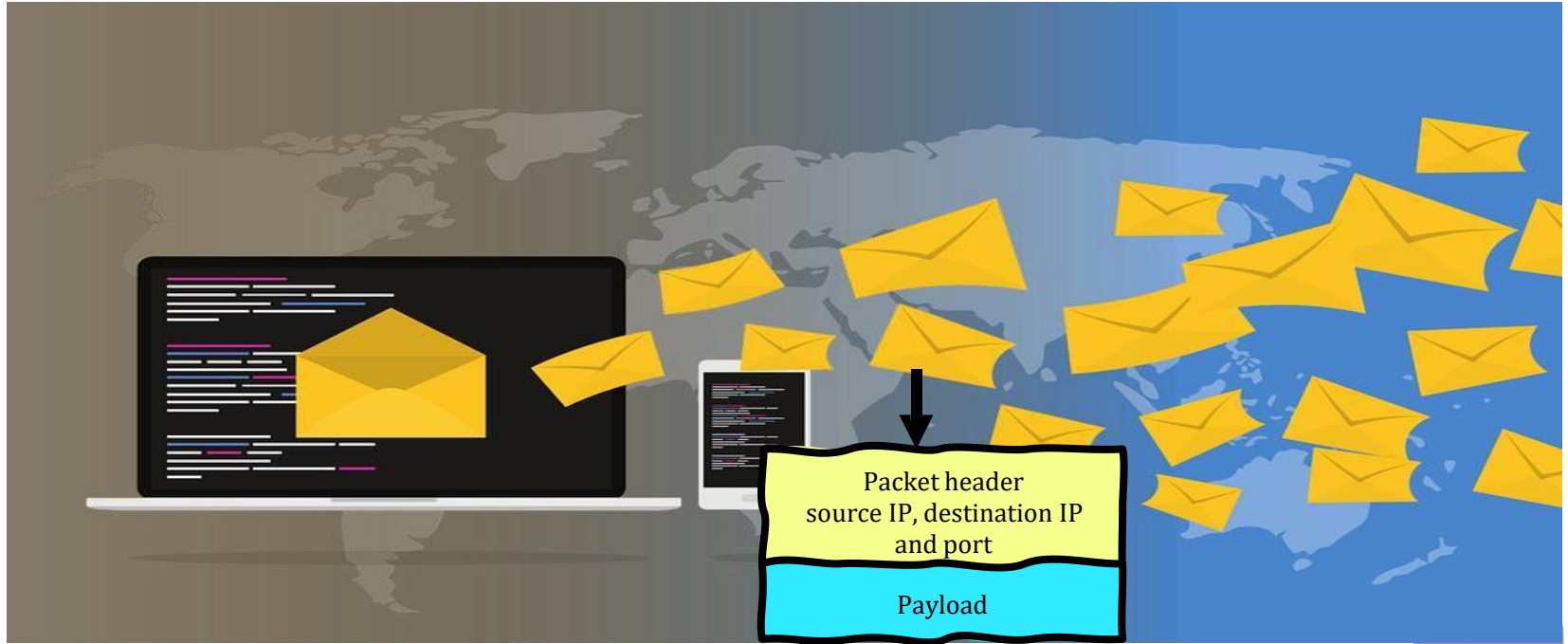
Firewall checks every single packet traversing the network. For example, when any user downloads a big-size file from the internet, the data of the file is divided into small chunks called packets. Each packet consists of two parts which are the packet header and packet payload.



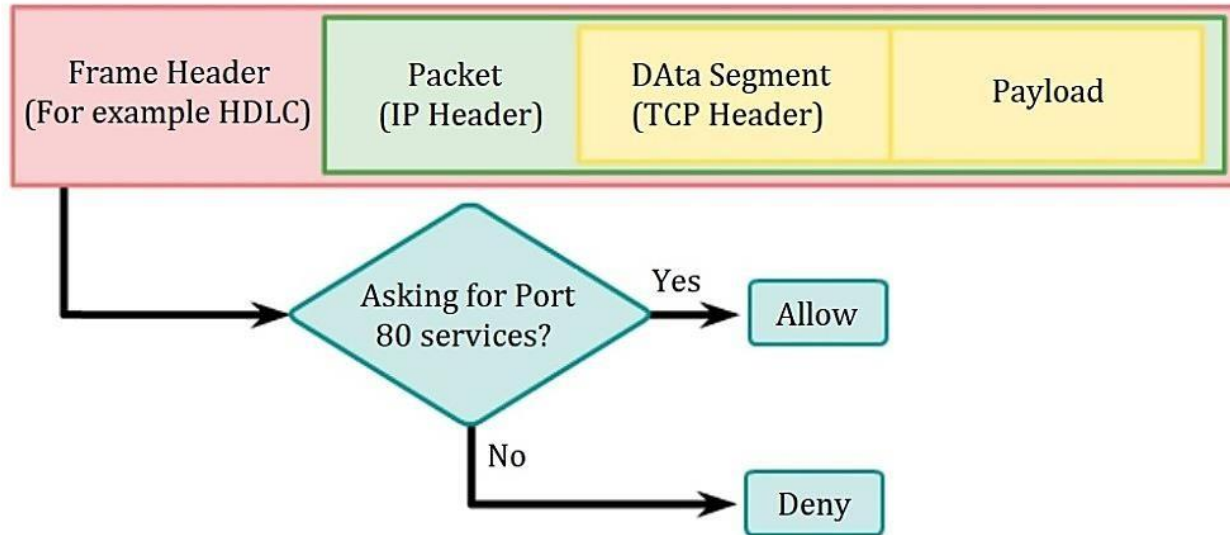
Packet header consists of info such as source and destination IP while the payload consists of the packet content



What does data packet mean?



Packet-filtering Firewalls(cont.)



Packet-filtering Firewalls Benefits

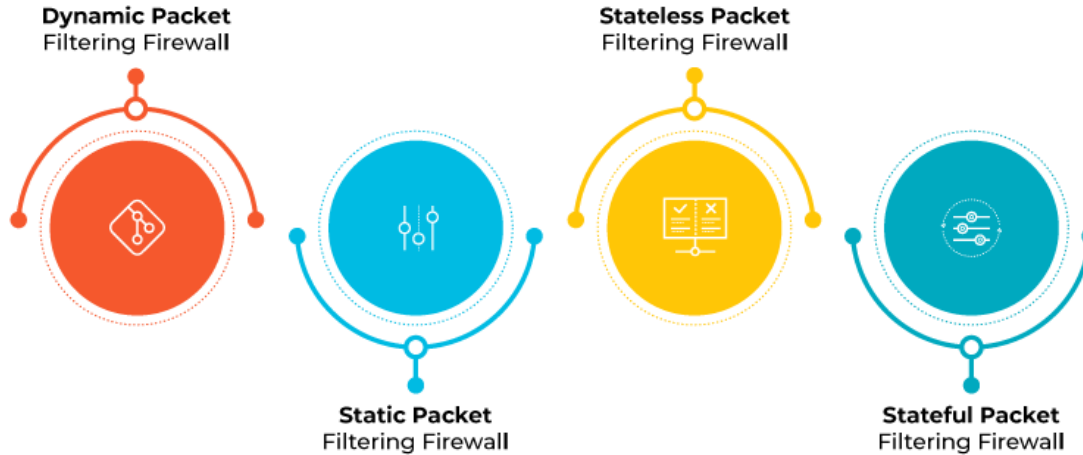
- High Speed Efficiency
- Transparent Operation
- Cost Efficiency
- Initial Simplicity and Ease of Use

Packet Filtering Firewall Challenges

- Limited Logging Capabilities
- Inflexibility
- Less Secure
- Stateless Operation
- Difficult to Manage
- Protocol Incompatibility

Types of Packet Filtering Firewalls

Types of Packet Filtering Firewalls



Types of Packet Filtering Firewalls

- Static Packet Filtering Firewall
- Dynamic Packet Filtering Firewall
- Stateless Packet Filtering Firewall
- Stateful Packet Filtering Firewall

Static Packet Filtering Firewall

- Uses fixed, manually defined rules.
- Rules remain unchanged until updated by an administrator.
- Operates at Network Layer (L3) — checks IP, port, protocol.
- Best for small, stable networks with predictable traffic.
- Advantages
 - Simple,
 - Low overhead,
 - Dependable
- Limitations
 - Not adaptive;
 - Manual updates required.

Dynamic Packet Filtering Firewall

- Adaptive rules that change based on traffic conditions.
- Can open or close ports dynamically as needed.
- Handles protocols with dynamic ports.
- Offers a balance between security and functionality.
- More responsive to current network state.

Stateless Packet Filtering Firewall

- Evaluates each packet in isolation.
- Makes decisions based on static rules (no session tracking).
- Fast and lightweight, minimal memory use.
- Suitable for low-traffic or low-risk environments.
- Lacks context — cannot detect multi-packet attacks.

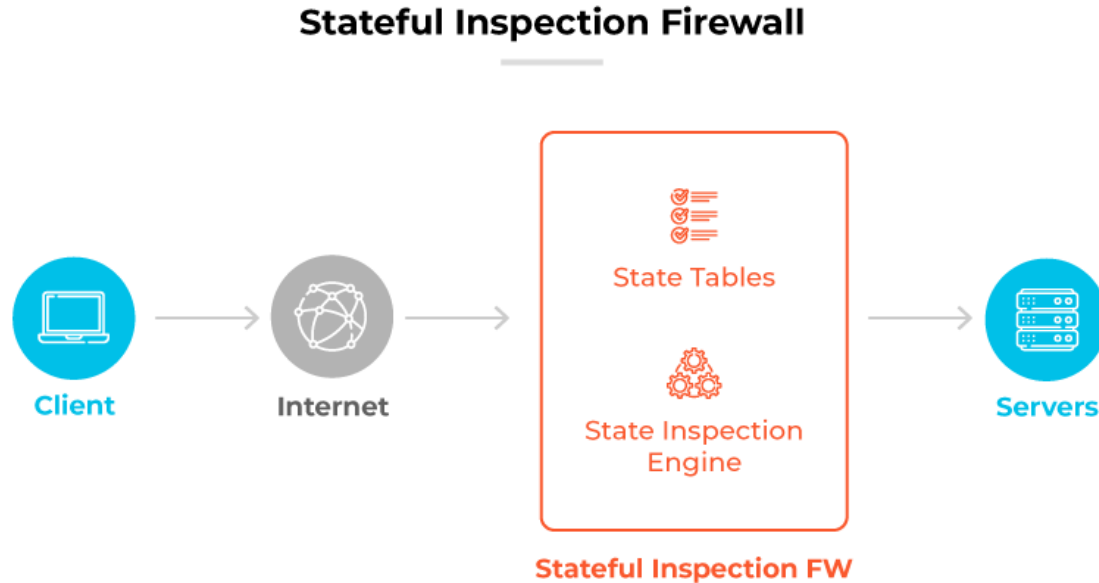
Stateful Packet Filtering Firewall

- Maintains a state table of active connections.
- Verifies that packets belong to valid, established sessions.
- Can detect and block unsolicited or spoofed packets.
- Provides stronger security than stateless firewalls.
- Foundation of most modern network firewalls.

Stateful Inspection firewalls.

- A network security device that monitors and maintains the context of active connections.
- Makes decisions about packets based on:
 - Predefined security rules, and
 - The state of current network sessions.
- Operates primarily at Layers 3 (Network) and 4 (Transport).
- Prevents unauthorized access by allowing only packets from valid, established connections.
- Stateful firewalls maintain tables containing information on each active connection, including the IP addresses, ports, and sequence numbers of packets.

Stateful Inspection Firewalls. (Cont.)



Stateful Inspection Firewalls. (Cont.)

Source Address	Source Port	Destination Address	Destination Port	Connection State
192.168.1.100	1030	210.22.88.29	80	Established
192.168.1.102	1031	216.32.42.123	80	Established
192.168.1.101	1033	173.66.32.122	25	Established
192.168.1.106	1035	177.231.32.12	79	Established
223.43.21.231	1990	192.168.1.6	80	Established
2122.22.123.32	2112	192.168.1.6	80	Established
210.922.212.18	3321	192.168.1.6	80	Established
24.102.32.23	1025	192.168.1.6	80	Established
223.21.22.12	1046	192.168.1.6	80	Established

How Do Stateful Firewalls Work?

- Connection Initiation:
 - Client sends request (e.g., TCP SYN).
 - Firewall records connection details in the state table.
- State Tracking:
 - Firewall monitors session status (open, established, closed).
 - Each packet is compared against the state table.
- Packet Inspection:
 - If part of a valid session → Allowed
 - If not recognized → Blocked or evaluated by rules
- Dynamic Session Updates:
 - State table continually updated as traffic flows.
 - Supports both TCP and connectionless protocols (UDP).
- Connection Termination:
 - Firewall validates and closes sessions securely.

Stateful Firewall Benefits

Stateful Firewall Benefits



Comprehensive
Traffic
Analysis



Dynamic
Network
Adaptability



Granular
Traffic
Control



Efficient
Resource
Utilization

Stateful Firewall Challenges

Stateful Firewall Challenges



Configuration
Complexity



Limitations in
Application Layer
Defense



Absence
of User
Authentication



Inadequate
Web Application
Security

Stateful vs. Stateless Firewalls

Stateful Firewalls

- Track ongoing connections
- Understand traffic context
- Remember previous interactions
- Detect patterns for security
- Manage complex traffic types
- Ideal for dynamic networks
- Provide detailed access control

Stateless Firewalls

- Inspect packets individually
 - Rely on set rules
 - No memory of past traffic
 - Block by basic criteria
 - Simple, uniform filtering
 - Suitable for basic needs
 - Offer general access control
-

Application/Proxy Firewall



Controls input, output, and access from/to an application or service.



Acts as an intermediary by impersonating the intended recipient.

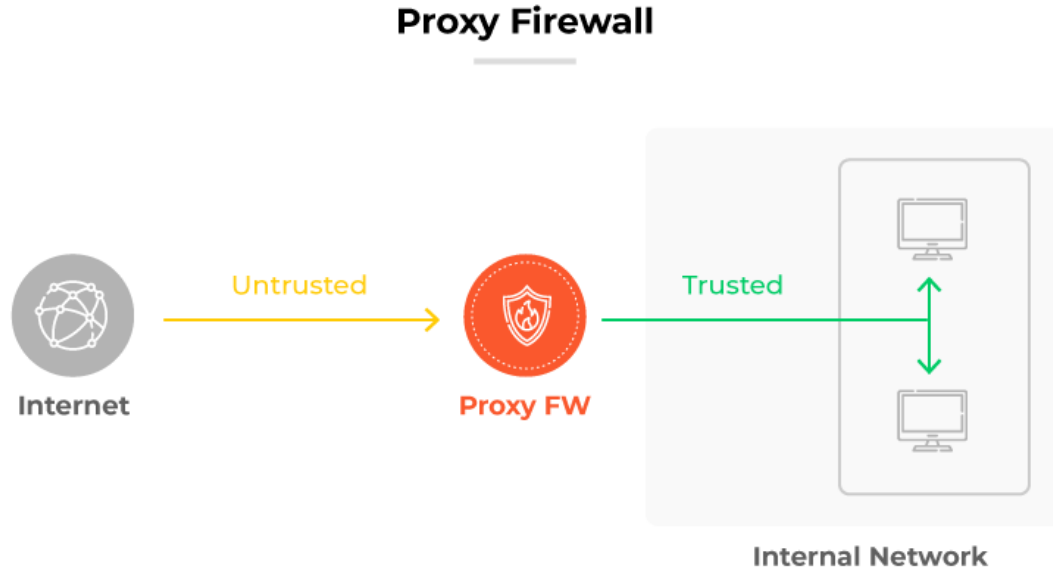


The client's connection terminates at the proxy and a separate connection is initiated from the proxy to the destination host.



Data on the connection is analyzed up to the application layer to determine if the packet should be allowed or rejected.

How Do Proxy Firewalls Work?



How Do Proxy Firewalls Work?

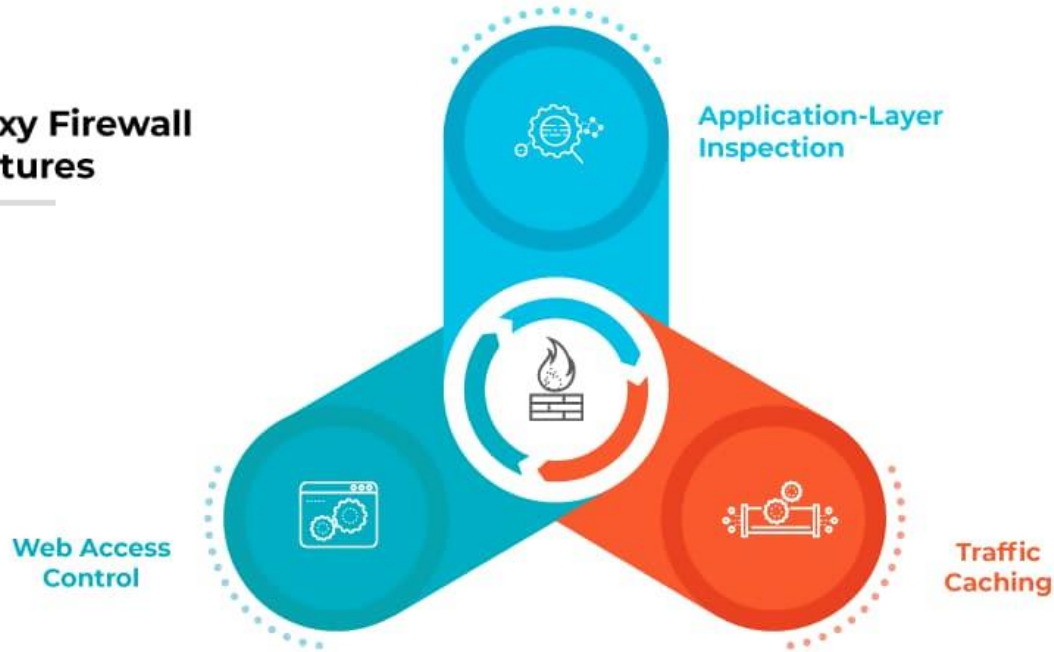
- Request Interception:
 - When an internal user tries to access a website, the request doesn't go directly to the internet.
 - Instead, it's sent to the proxy firewall, which acts as an intermediary or 'middleman.'
- Evaluation & Filtering:
 - The firewall inspects the request content — not just IP addresses and ports, but actual data payloads.
 - DPI allows detection of malware, unauthorized file transfers, or policy violations.
- Connection Mediation:
 - Once the proxy firewall approves the request, it creates a new outbound connection to the internet on behalf of the user.
 - The external server never directly interacts with the internal client — this breaks the direct link and hides internal addresses..

How Do Proxy Firewalls Work?

- Response Handling:
 - The response from the internet is received by the proxy, inspected again for threats, and then safely delivered to the client.
 - This ensures that the internal network never directly trusts external data.
- Additional Capabilities:
 - Caching: frequently accessed content is stored locally to improve speed.
 - Compression: reduces bandwidth consumption.
 - Filtering: administrators can block certain sites or services.
 - Because it works at Layer 7, it can prevent sophisticated phishing, malware, or data exfiltration attempts that bypass lower-level defenses.

Proxy Firewall Features

Proxy Firewall Features



Proxy Firewall Benefits

Proxy Firewall Features



Enhanced
Security



Improved
Privacy



Granular
Control



Advanced Threat
Detection



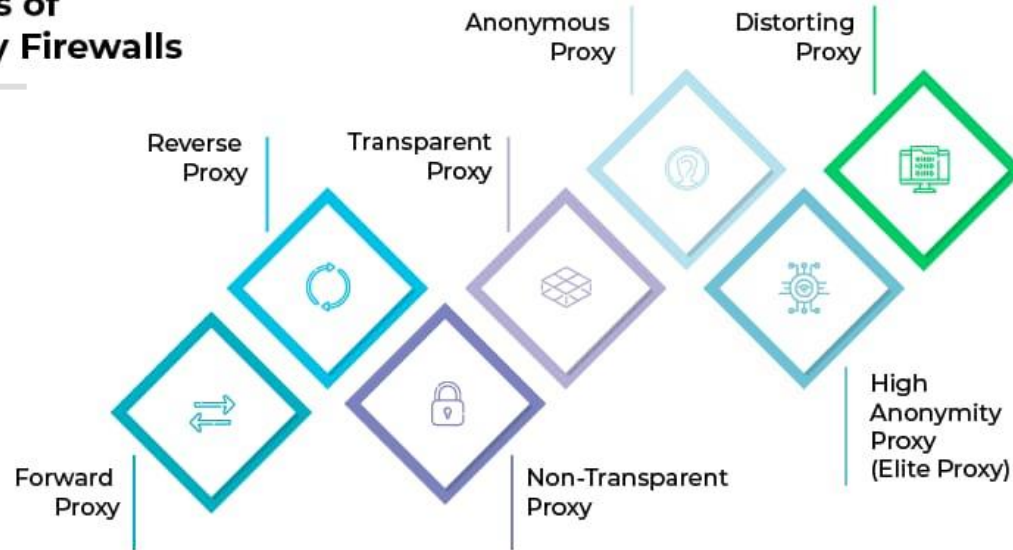
Network Performance
Optimization

Proxy Firewall Challenges

- Managing Network Performance
- Protocol Limitations
- System Complexity
- Possible Single Point of Failure
- Encryption and Privacy Concerns

Types of Proxy Firewalls

Types of Proxy Firewalls



Web Application Firewall (WAF)

- A Web Application Firewall protects web applications by monitoring, filtering, and blocking HTTP/S traffic between a web application and the internet.
- Operates at the Application Layer (Layer 7).
- Inspects the content of web requests (URLs, forms, cookies, headers).
- Detects and blocks attacks that exploit application vulnerabilities, such as:
 - SQL Injection (SQLi)
 - Cross-Site Scripting (XSS)
- Can be network-based, host-based, or cloud-based.
- Uses rule sets and behavioral analysis to distinguish between legitimate and malicious traffic.

Next-generation firewalls (NGFWs).

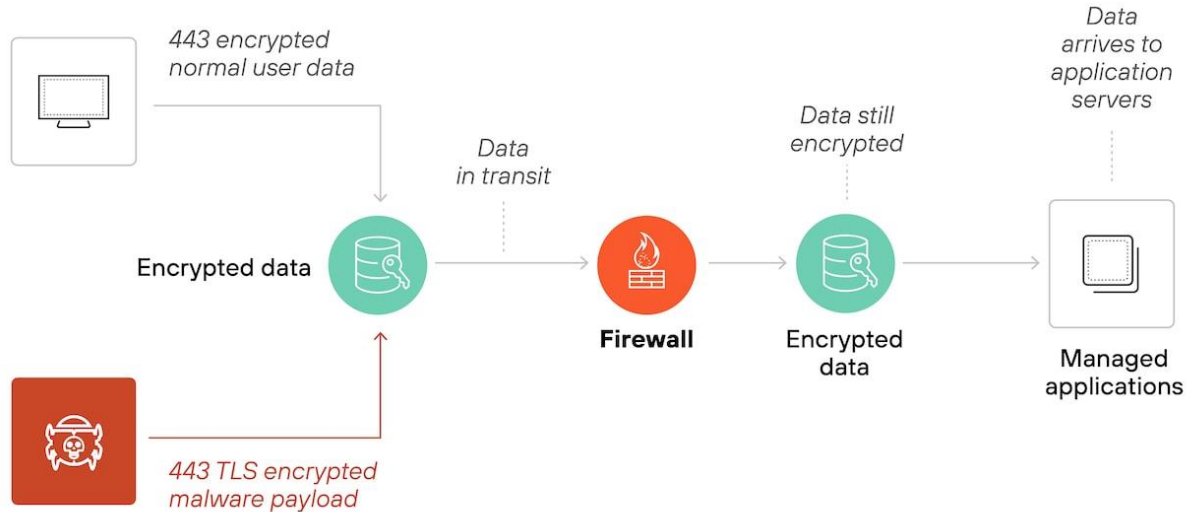
- A next-generation firewall (NGFW) is a type of firewall that provides capabilities beyond a traditional, stateful firewall.
- While a traditional firewall typically provides stateful inspection of incoming and outgoing network traffic, a next-generation firewall includes additional features like application awareness and control, integrated intrusion prevention, and cloud-delivered threat intelligence.

What created the need for NGFWs?

- Applications started behaving differently. Many used the same ports. Or tunneled through SSL. This made it harder for traditional firewalls to tell good from bad—or even know which app was running.
- At the same time, threats got more evasive. Malware could hide in encrypted traffic. Attacks targeted specific apps, users, and data.
- Traditional tools couldn't keep up. They couldn't inspect encrypted content. And they couldn't tie traffic back to users in meaningful ways.

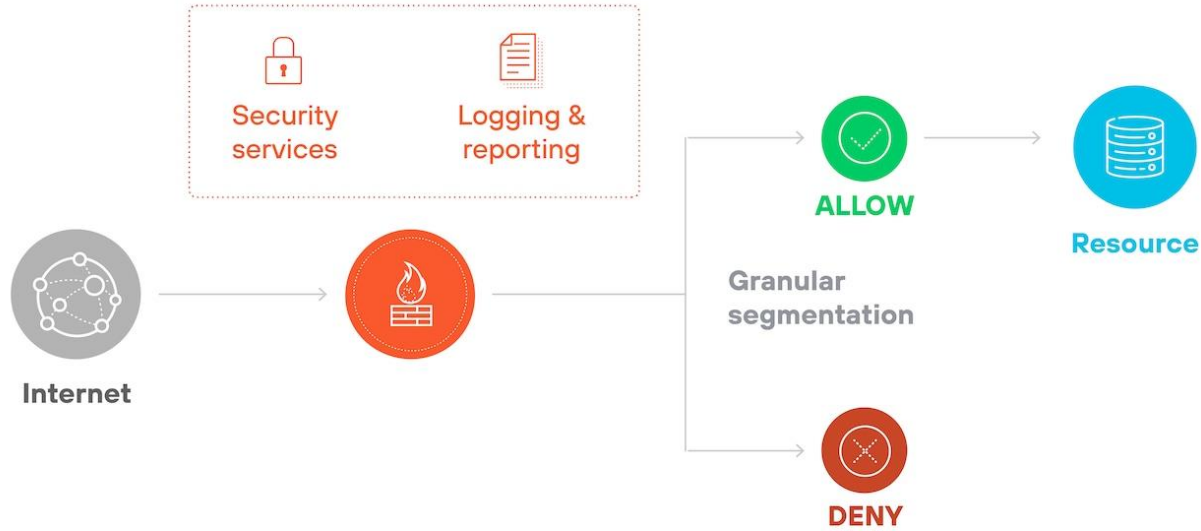
Next-generation firewalls (NGFWs).

Traditional network security gaps in a modern threat environment



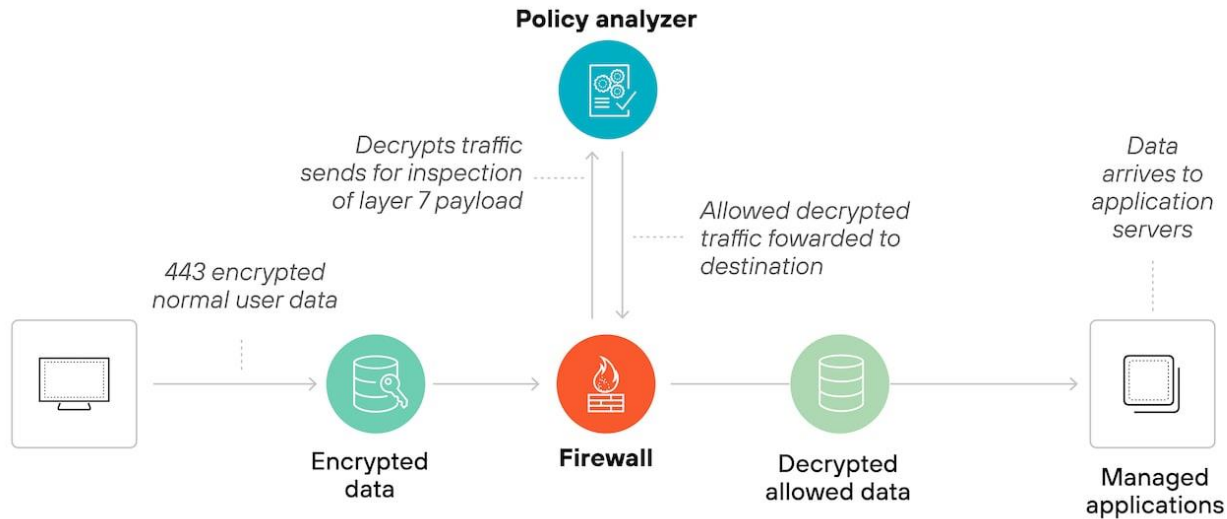
How does an NGFW work?

Next-generation firewall (NGFW)



How does an NGFW work?

Layer 7 application inspection and policy enforcement



What is Unified Threat Management (UTM)?

- Unified Threat Management (UTM) is an all-in-one security mechanism that integrates various features and operations of security into one appliance.
- UTM strives to enhance the efficiency of managing security technologies.
- UTM is composed of several security technologies, such as
 - firewall,
 - IDPS,
 - antivirus/antimalware,
 - content/spam filter,
 - and VPN support.

Thank you

